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The optimization of low Earth orbit satellite constellation visibility with genetic algorithm for improved navigation potential

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To tackle visibility optimization challenges in Low Earth Orbit satellite constellations, this study proposed an enhanced framework based on an adaptive parallel Genetic Algorithm (GA). The aim was to improve both navigation accuracy and dynamic robustness. A hybrid constellation was designed by integrating polar, Walker, and Orthogonal Circular Orbits, with a dynamic relaxation factor, γ, explicitly included to compensate for J2 perturbation effects. The framework introduced three key innovations: (1) an adaptive parameter adjustment mechanism guided by population diversity entropy, (2) a parallel fitness evaluation strategy optimized for multi-core architectures, and (3) a simplified yet structured fitness function design. Experimental results showed that in a standard 100-satellite scenario, the optimized constellation achieved an average of 14.3 visible satellites—3.6% better than D-NSDE. The Position Dilution of Precision (PDOP) was reduced to 2.3, and global coverage reached 95.6%. After a single satellite failure, coverage dropped by only 3.5%, and remained at 94.8% after 10 years of orbital perturbation. Even in ultra-large-scale scenarios with 500 and 1,000 satellites, the framework maintained PDOP values of ≤ 2.8, with convergence times under 210 s. Overall, the proposed GA outperformed Particle Swarm Optimization and Dynamic Non-dominated Sorting Differential Evolution in visibility, robustness, and computational efficiency. This provides a crucial technical foundation for next-generation Global Navigation Satellite System enhancements.

Keywords Genetic algorithm, Low earth orbit satellite constellation, Constellation optimization, Navigation precision, Satellite visibility

The rapid advancement of technology and increasing demand for applications have raised expectations for accurate satellite positioning in navigation systems[1–3]. Global Navigation Satellite System (GNSS) has become critical infrastructure across many sectors, including aviation, transportation, defense, and civilian applications. However, their performance remains limited in challenging environments such as urban canyons with signal blockages or regions affected by strong electromagnetic interference. Issues with accuracy, integrity, and anti-jamming capabilities continue to pose significant challenges[4–6]. To overcome these limitations, integrating Low Earth Orbit (LEO) satellite constellations with traditional Medium Earth Orbit (MEO)-based GNSS has emerged as a promising enhancement strategy[7–9]. LEO constellations offer several advantages. Their rapid orbital movement causes dynamic changes in satellite geometry, enabling more flexible and continuous regional coverage. This is especially beneficial in high-latitude areas and regions with complex terrain, where service availability often suffers[10]. Additionally, the closer proximity of LEO satellites to Earth provides stronger signal reception and shorter transmission delays. These factors improve positioning accuracy, accelerate Precise Point Positioning (PPP) convergence, and enhance overall system reliability[11]. Consequently, LEO constellations complement traditional GNSS and support the development of next-generation integrated navigation and positioning solutions.

Within this context, optimizing LEO constellation configurations to maximize satellite visibility—meaning the number of satellites simultaneously observable—and minimize Position Dilution of Precision (PDOP) has become a central research focus. The objective is to boost both precision and robustness of navigation systems. To address this complex multi-objective optimization (MOO) problem, various methods have been explored recently. Early studies primarily employed metaheuristic algorithms such as Genetic Algorithms (GA), Particle

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Swarm Optimization (PSO), and Simulated Annealing (SA)[12–15]. While these approaches showed initial promise, scaling them to large constellation designs revealed several challenges. These include high computational costs leading to long optimization times, a tendency to get stuck in local optima causing premature convergence, and limited ability to handle complex physical and operational constraints, such as orbital perturbations, safety margins, and fault tolerance[16].

To address these challenges, more advanced and intelligent optimization strategies have been developed. Recent progress generally falls into four main categories: (1) Mathematical Programming Techniques: Methods like Dynamic Programming (DP) have shown effectiveness for specific subproblems, such as ground station placement and energy-efficient orbital maneuvers. However, they face difficulties handling the combinatorial complexity of high-dimensional, mixed discrete-continuous constellation design problems[17]. (2) Enhanced Evolutionary Algorithms: Approaches such as Dynamic Non-dominated Sorting Differential Evolution (D-NSDE) have improved convergence speed and the exploration of the Pareto front. Despite this, their capacity to manage dynamic constraints—like long-term orbital perturbations—remains insufficiently explored[18]. (3) Hybrid Metaheuristic Algorithms: Combinations like GA-PSO aim to balance exploration and exploitation. However, these methods often require complex parameter tuning, which increases configuration effort and computational cost[19]. (4) Learning-Based Methods: Reinforcement Learning (RL) holds promise for online adaptation and dynamic optimization. Yet, it typically requires extensive simulation or real-world data for effective training. Its generalizability and stability across diverse application scenarios remain open challenges[20].

Despite significant progress, several key challenges remain: (a) Efficiently exploring and achieving global convergence in high-dimensional, mixed-integer parameter spaces remains difficult. These spaces include variables such as the number of orbital planes, satellites per plane, inclination angles, phase factors, and orbital altitudes. This challenge is especially pronounced when designing complex hybrid constellations that combine Walker, polar, and equatorial orbits. (b) Developing constellations inherently robust to dynamic perturbations—such as orbital plane precession and phase drift caused by Earth's J2 gravitational effect—and tolerant of satellite failures requires an optimization framework that explicitly incorporates these uncertainties during design. Relying on post hoc analysis is insufficient. (c) Achieving effective trade-offs among multiple, often conflicting objectives is challenging. These objectives include satellite visibility, PDOP, deployment and maintenance costs, coverage stability, and service guarantees in priority regions. Addressing this complexity requires powerful MOO techniques. These must generate high-quality, well-distributed Pareto solutions under complex constraints.

To address these challenges—particularly the optimization of hybrid LEO constellations in high-dimensional parameter spaces with robustness to orbital disturbances and system faults—this study proposes an enhanced GA framework. While traditional methods such as standard GA, PSO, and SA have laid important groundwork, they often fall short in highly complex scenarios. These methods tend to rely on static parameters, perform inefficiently in large-scale or high-fidelity simulations, and struggle to capture the coupling between orbital parameters and dynamic physical constraints effectively.

The proposed enhanced GA framework introduces three key innovations: (1) Adaptive Parameter Tuning: Crossover and mutation rates are dynamically adjusted according to the evolutionary stage and population diversity. This balances global exploration with local exploitation, improving performance in constrained and dynamic environments. (2) Parallel Fitness Evaluation: A parallel computing strategy utilizes multi-core architectures to speed up fitness evaluations. This approach scales efficiently to large constellations and long simulation times. (3) Structured Fitness Function Optimization: Geometric preprocessing and caching techniques are employed to significantly reduce the computational load of fitness calculations. These innovations are tightly integrated with a hybrid constellation coverage model. This model incorporates geometric relationships and a dynamic relaxation factor designed to mitigate the effects of orbital perturbations.

The study aims not only to apply a GA-based method but also to demonstrate that the enhanced framework offers clear advantages in managing the complexity and robustness demands of hybrid LEO constellation design. Comparative evaluations are performed against conventional metaheuristics (PSO, SA) and a Dynamic Programming (D-NSDE), using metrics focused on optimization performance, computational efficiency, and, most importantly, robustness.

This study makes the following main contributions: (1) Robust Hybrid Constellation Design for Dynamic Environments: A hybrid LEO constellation framework is proposed, integrating multiple orbital types—polar, Walker, and Orthogonal Circular Orbits (OCO). By incorporating a dynamic relaxation factor and other mechanisms, the framework enhances structural resilience and robustness against orbital variations such as J2 perturbations. (2) Advanced GA for Complex Constrained MOO: An enhanced GA framework is developed featuring three key components: (i) an adaptive control mechanism for genetic operators based on evolutionary feedback, (ii) a parallel fitness evaluation strategy optimized for multi-core architectures, and (iii) a simplified yet structurally informed fitness function designed for hybrid constellation optimization with multiple objectives and constraints. 3. Comprehensive Performance and Robustness Validation: The proposed method is systematically evaluated against mainstream optimization algorithms. Beyond standard performance metrics, the assessment includes robustness criteria like stability under orbital perturbations and resilience to satellite failures. Results show that the GA significantly outperforms alternatives in maintaining navigation capability.

In summary, this study presents a high-performance, dynamically robust GA-based optimization framework tailored for hybrid LEO constellations. The findings offer key technical support for next-generation GNSS enhancements. Paper Structure: Sect. 2 reviews related work, Sect. 3 details the constellation model and GA framework, Sect. 4 describes the experimental setup and discusses results, and Sect. 5 concludes and outlines future research directions.

Literature review

In recent years, researchers have developed various optimization methods for satellite constellation design. These methods fall into several broad categories, each with its own strengths and limitations in addressing challenges such as dynamic robustness (including orbital perturbation compensation and fault tolerance), efficient search in high-dimensional mixed-parameter spaces, and complex multi-objective trade-offs.

Mathematical programming methods aim to find optimal solutions through formal mathematical models. DP, for example, has proven effective for sequential decision problems with optimal substructure and overlapping subproblems. It has been applied to tasks like constellation scheduling, resource allocation, and partial parameter tuning. For instance, Wang et al. (2024) proposed a DP-based navigation resource allocation algorithm that efficiently handled specific subproblems in LEO constellation design by managing discrete decision sequences and resource constraints[17]. However, DP faces major limitations when applied to the core problem of optimizing large-scale hybrid constellations, which involve both continuous variables (such as orbital inclination and RAAN) and discrete variables (such as the number of satellites and planes). DP suffers from the "curse of dimensionality," where computational complexity grows exponentially with the number of variables. This makes it impractical for full-scale, high-dimensional constellation design. Moreover, embedding dynamic perturbation compensation or fault recovery mechanisms within DP models remains a significant challenge. Similarly, Integer Linear Programming (ILP) and Mixed-Integer Linear Programming (MILP) have been employed for coverage optimization but depend heavily on linearization. This limits their ability to capture nonlinear orbital dynamics and geometric constraints, reducing their usefulness in high-fidelity, multi-objective scenarios.

Improved evolutionary computation methods, a major class of metaheuristic algorithms, have gained popularity for constellation optimization due to their global search abilities and flexibility in handling nonlinear problems. Common examples include GA, PSO, and Differential Evolution (DE), often enhanced with various strategies to boost performance. For example, Wang et al. (2025) proposed a multi-layer LEO constellation optimization method based on D-NSDE. This method incorporated dynamic adjustment and non-dominated sorting to enhance Pareto front exploration, achieving promising results for medium-scale constellations[18]. Despite these advancements, most evolutionary algorithms still face difficulties managing strong dynamic constraints. Their parameter control strategies often do not explicitly consider long-term orbital disturbances, such as orbital plane precession caused by the J2 effect, which can undermine constellation stability. Additionally, they typically lack mechanisms for rapid and autonomous reconfiguration when satellite failures occur. Although algorithms like D-NSDE have improved search efficiency in high-dimensional mixed-parameter spaces, convergence speed and computational resource demands remain critical bottlenecks, especially for large-scale or highly heterogeneous constellation designs. Producing well-distributed, high-quality Pareto fronts in complex multi-objective problems also remains a significant challenge. SA has been applied to constellation design as well. By probabilistically accepting worse solutions, SA attempts to escape local optima. However, it often performs inefficiently in high-dimensional, multi-objective scenarios and lacks a systematic framework to incorporate dynamic robustness[19].

Hybrid heuristic and metaheuristic methods aim to combine the strengths of different algorithms to overcome limitations of single approaches. Common strategies include integrating GA with PSO or embedding local search operators within global optimization frameworks. For example, Huang et al. (2021) explored GA-based designs for integrated communication and navigation LEO constellations, highlighting the potential of hybrid heuristics[20]. More generally, GA-PSO hybrids use GA's selection and crossover for global exploration, paired with PSO's velocity updates for local exploitation, striving to balance convergence speed with solution quality. However, hybrid methods often increase algorithmic complexity and introduce additional parameter tuning challenges. More importantly, existing hybrids rarely focus on dynamic robustness, which is central to this study. Their core operators—such as crossover, mutation, and particle velocity updates—do not inherently model orbital perturbations or include fault recovery mechanisms. As a result, these methods struggle to produce constellation configurations that resist perturbations or can be easily reconfigured after satellite failures. While hybrids can be more robust than single algorithms for ultra-high-dimensional mixed-parameter spaces and complex multi-objective problems, their computational cost and reliability in consistently finding global optima remain uncertain. Further validation is necessary in complex constellation design contexts. Other approaches, like Grey Wolf Optimization (GWO) and its variants, have been applied to satellite mission planning. Yet, their social hierarchy-based update mechanisms are often criticized for poor search efficiency in high-dimensional, mixed discrete-continuous spaces.

Hong et al. (2023) simulated 180 globally distributed LEO satellites and found that incorporating LEO observations notably improved the convergence speed of PPP, particularly for single-GNSS systems. Their study also showed that LEO satellites enhanced ambiguity resolution success rates and real-time dynamic positioning performance, while significantly reducing positioning time[21]. The key methods, findings, and limitations of these related studies are summarized in Table 1.

Significant advancements have been made in the field of satellite constellation optimization. For example, Niccolai et al.[27] proposed an optimization method for ground station deployment based on evolutionary algorithms[22], but their study did not explore the impact of dynamic parameter adjustment on MOO. Wu et al.[28] applied an improved Grey Wolf Optimizer (GWO) to solve satellite mission planning problems[23], yet they did not investigate the acceleration effects of parallel computing for large-scale constellation optimization. Huang et al.[29] introduced a fitness feedback mechanism into the DE algorithm[24], though the stability of their method under complex constraints remains to be verified.

The field of satellite constellation optimization has made continuous progress, spanning rigorous mathematical programming, flexible metaheuristics and their hybrid variants, as well as emerging learning-based approaches. Despite these advances, significant challenges remain. The task of designing hybrid LEO constellations with dynamic robustness remains challenging. Such constellations must resist orbital perturbations and tolerate

Research authors	Research topic	Research method	Research findings	Limitations
Wang et al.[17]	LEO Satellite Navigation Resource Allocation Based on DP	DP	Efficiently solved subproblems such as ground station deployment and task scheduling	Limited applicability due to combinatorial explosion in high-dimensional mixed continuous-discrete design problems
Wang et al.[18]	Multi-layer LEO Constellation Optimization Using D-NSDE	D-NSDE	Demonstrated good convergence and Pareto front exploration in MOO	Insufficient adaptation to strong dynamic constraints such as orbital perturbations
Tao et al.[19]	Joint Optimization of Altitude and Beamwidth for IoT-Oriented LEO Constellations	Hybrid optimization combining Genetic Algorithm and multi-objective heuristics	Improved system capacity and coverage effectiveness in IoT scenarios	Complex parameter tuning; algorithm stability dependent on initial settings
Huang et al.[20]	Optimal Design of Communication and Navigation Integrated LEO Constellations Using Genetic Algorithms	GA	Proposed an optimization strategy integrating communication and navigation	Lacked comprehensive consideration of dynamic factors like orbital perturbations and constellation robustness
Hong et al.[21]	Impact of Low-Earth Orbit Satellites on PPP Convergence Speed	Simulated the effect of 180 globally distributed LEO satellites on PPP convergence	Found that LEO satellites significantly improved PPP convergence speed, especially in single GNSS scenarios	Differences between simulation results and real-world applications require further investigation

Table 1. Key contents, methods, research findings, and limitations of related studies.

faults. This design problem occurs within high-dimensional mixed-parameter spaces. Additionally, it requires deep MOO. This involves balancing performance, cost, and robustness. To date, this complex problem has not been fully addressed. Mathematical programming methods face limitations due to dimensionality and model complexity. Evolutionary algorithms, such as D-NSDE, struggle to embed dynamic robustness effectively and to scale efficiently for ultra-large problems. Hybrid heuristic methods often increase complexity but lack targeted solutions for these core issues. Meanwhile, learning-based approaches like RL incur high computational costs and remain immature for comprehensive constellation design. These challenges remain unresolved. They include efficiently and explicitly integrating dynamic perturbation compensation and fault recovery into the optimization process. Additionally, thorough exploration of the multi-objective Pareto front in high-dimensional mixed spaces is required. Addressing these issues forms the main motivation for this research. The adaptive parallel GA framework proposed here is specifically designed to address these critical problems.

Research model

Configuration analysis of the single constellation

The mixed constellation analyzed in this study is composed of three main types: polar, Walker, and OCO constellations. The model construction places special emphasis on active compensation for orbital dynamics perturbations, primarily the nodal precession and phase drift caused by the Earth's non-spherical gravitational field (the J2 term). This active compensation is key to achieving dynamic robustness.

Polar orbit constellation

A polar orbit constellation is a type of LEO satellite constellation where the orbital planes pass near the Earth's poles. This design allows satellites to achieve global coverage during each orbit, making it particularly suitable for providing service in polar regions. For example, Hasbi et al. (2020) evaluated equatorial and polar tracking capabilities using polar-orbiting satellites. Their study showed that polar constellations achieved significantly higher detection probabilities in high-latitude areas compared to equatorial constellations[25]. These findings support both the definition of the coverage belt semi-width in Eq. (3) and the applicability of polar constellations for high-latitude coverage. Ma et al.[31] proposed a hybrid constellation design method based on a GA[26]. Their MOO framework provided a theoretical foundation for the weighted objective function used in Eq. (15). Similarly, Liu et al.[32] analyzed the influence of orbital parameters on LEO occultation events[27], confirming the relationship between the number of orbital planes and the coverage belt width as expressed in Eq. (6).

Typically, satellites in polar orbits follow near-circular or elliptical paths, which increases the frequency of satellite passes over high-latitude regions and enhances overall coverage density. Zhang et al.[33] introduced a multi-GNSS availability assessment metric specifically for polar regions[28]. Kleinboehl et al.[34] further demonstrated a strong correlation between the spacing of adjacent orbital planes (Δ_1) and the accuracy of atmospheric parameter inversion, using vertical profiles of greenhouse gases retrieved from a polar-orbiting satellite constellation[29]. This finding offers empirical evidence on a planetary scale, reinforcing the general applicability of the geometric coverage model proposed in this study. Pontani et al.[35] also investigated deployment strategies for polar ice monitoring satellite constellations[30]. Collectively, these studies validate the conclusion presented in "Configuration analysis of the single constellation" regarding the "enhanced high-latitude coverage density" achieved by polar orbit constellations. To define the semi-geocentric angle θ for a single satellite's coverage area, the following assumptions are made: The Earth is modeled as a perfect sphere; The satellite's orbital altitude is denoted by h; The minimum elevation angle is E_{min}; The Earth's radius is R_0. Based on the law of sines, the geometric relationship at the boundary of the satellite coverage area is given by Eq. (1):

$$\frac{R_0}{\sin[\pi - \theta - (\frac{\pi}{2} + E_{min})]} = \frac{R_0 + h}{\sin(\frac{\pi}{2} + E_{min})} \tag{1}$$

By applying trigonometric identities, this leads to the following simplified expression for θ:

$$\theta = \arccos\left(\frac{R_0}{R_0 + h} * cosE_{min}\right) - E_{min} \tag{2}$$

This derivation clarifies the physical meaning of θ, which represents the satellite's coverage cone angle. The value of θ is determined jointly by the satellite's orbital altitude and the minimum elevation angle. For polar orbit constellations, the semi-width (b) of a single coverage belt is related to the semi-geocentric angle (θ) and the number of satellites (S) in a polar orbital plane. The semi-width b can be derived using spherical right triangle geometry. Assuming that satellites are evenly distributed within the orbital plane, the phase difference between two adjacent satellites is π/S. When a ground point P lies at the edge of a coverage belt, the line connecting the point P and the corresponding satellite forms a projected angle of π/S within the orbital plane. A spherical right triangle OPQ is constructed, where O is the geocenter, Q is the satellite position, and PQ corresponds to the coverage belt semi-width b. The side OQ represents the semi-geocentric angle θ, and the hypotenuse OP corresponds to the phase difference π/S. The relationship between b and θ can be further established using the spherical law of cosines, as shown in Eq. (3):

$$\cos\left(\frac{\pi}{S}\right) = \cos\theta \bullet \cos b \tag{3}$$

Equation (3) shows that the phase difference between adjacent satellites ($\frac{\pi}{S}$) directly affects the coverage belt width. Moreover, Eqs. (5) and (6) introduce relaxation factors (γ) to quantify the longitudinal spacing requirements between prograde and retrograde orbital planes. By progressively linking geometric constraints with orbital parameters, a complete mathematical model is constructed, describing the transition from single-satellite coverage to multilayer global coverage. In this study, it is assumed that the phase difference between adjacent satellites remains constant at π/S. Although this assumption simplifies the construction of continuous coverage belts, it does not fully match real-world orbital dynamics. In practice, perturbations such as Earth's non-spherical gravity and atmospheric drag gradually alter orbital elements, leading to phase drift over time. For example, Hippelheuser and Elgohary (2021) reported that the nodal precession rate of LEO satellites could reach several degrees per day. Over time, such effects may significantly disrupt the initial phase relationships. To address this issue, a relaxation factor γ ($0 < \gamma \leq 1$) is introduced in this study. This factor allows dynamic adjustment of orbital plane spacing, helping to maintain continuous coverage despite the effects of orbital perturbations.

The mathematical expression for the coverage belt semi-width is provided in Eq. (4):

$$b = \arccos\left(\frac{cos\theta}{cos\frac{\pi}{S}}\right) \tag{4}$$

Satellites in polar orbit constellations, located on adjacent orbital planes, exhibit two types of relative motion: anterograde and retrograde. For instance, Zhang et al. (2021) designed a distributed MIMO system based on polar-orbiting LEO constellations[31]. Their channel capacity model provided a reference for constructing the geometric matrix in Eq. (11). Similarly, Luu et al. (2021) analyzed on-orbit service constraints for LEO constellations, highlighting the effect of orbital altitude on coverage range[32]. Their findings supported the altitude selection rationale for OCO constellations. The derivation presented in this study is based on the following assumptions:

(1) The Earth is modeled as a perfect sphere, ignoring its oblate shape and surface terrain variations.
(2) Satellite orbits are assumed to maintain fixed circular altitudes.
(3) The phase difference between adjacent orbital planes remains constant at π/S, ensuring the continuity of coverage belts.
(4) The coverage belt boundaries are determined through spherical geometric relationships between the semi-geocentric angle θ and the semi-width b.

The circular orbit and idealized spherical Earth assumptions used in this study simplify the coverage modeling, but their applicability needs to be clarified: (1) Communication tasks: For LEO satellites, the orbital period is relatively short, and the circular orbit assumption effectively characterizes continuous coverage. (2) Earth observation: When polar constellations use circular orbits, the revisit period stabilizes at 7 days (with an error of ± 1.2 h). However, the Earth's oblateness (the J_2 term) causes the orbital plane to precess at a rate of 3.2° per day. In this study, the dynamic adjustment of orbital spacing through the relaxation factor γ (Eqs. 5 and 6) reduces the coverage blind spot area above 70° latitude, meeting the precision requirements for tasks such as meteorological monitoring. (3) High-altitude task limitations: For Medium Earth Orbit (MEO) or Geostationary Earth Orbit (GEO) satellites, the orbital perturbations caused by the Earth's non-spherical gravitational field significantly increase. In such cases, numerical orbit forecasting models are required to correct the coverage predictions.

Satellites on prograde orbital planes maintain a fixed phase difference, which ensures seamless coverage when arranged in a staggered pattern with a phase difference of $\frac{\pi}{S}$. This arrangement also increases the longitudinal difference, Δ_1, between adjacent prograde orbital planes, thus reducing the number of required orbital planes. The expression for Δ_1 is given in Eq. (5):

$$\Delta_1 = (\theta + b)\gamma \tag{5}$$

In Eq. (5), γ is a scaling factor with a value less than or equal to 1. This factor adjusts the spacing between orbital planes by allowing for flexibility in their arrangement. γ reflects practical requirements for coverage belt overlap

redundancy. When γ is close to 1, the spacing between orbital planes is maximized, which may reduce coverage continuity. On the other hand, decreasing γ results in a tighter spacing, requiring additional orbital planes to fill coverage gaps. This parameter balances the need for coverage continuity with the number of orbital planes by adjusting the spacing between adjacent planes. The optimal value of γ is determined through GA optimization to meet global single-layer coverage conditions. Since the phase difference between adjacent satellites changes over time, areas outside the coverage band cannot be fully utilized. Therefore, retrograde orbital planes must be positioned closer together to maintain continuous coverage. The longitudinal difference Δ_2 between retrograde orbital planes is given in Eq. (6):

$$\Delta_2 = 2b\gamma \tag{6}$$

Equation (6) shows that retrograde orbital spacing is half that of prograde orbital spacing. This ensures a denser overlap of adjacent coverage belts, especially in high-latitude regions. To achieve global single-layer coverage in a polar orbit constellation, the following conditions must be met:

$$(P_P - 1)\,\Delta_1 + \Delta_2 = [(P_P - 1)\,\theta + (P_P + 1)\,b]\,\gamma = \pi \tag{7}$$

In Eq. (7), P_P represents the number of polar orbital planes. By combining Eqs. (5) and (6), the global single-layer coverage condition for a polar orbit constellation is established. The sum of the angular separations between P_P prograde orbital planes and one retrograde orbital plane must equal the full longitude coverage condition, π. This leads to a closed-form equation involving P_P, θ, b, and γ. This equation serves as the mathematical foundation for the subsequent optimization of mixed constellation parameters using a GA. To validate Eq. (7), a practical polar orbit constellation deployment is considered with the following parameters: orbital altitude (h) = 500 km, number of satellites (S) = 10, semi-geocentric angle (θ) = 26.9°, coverage belt semi-width (b) = 40.1°, and scaling factor (γ) = 0.95. Substituting these values into Eq. (7), the experiment gets the equation: $[(P_P - 1)\,\theta + (P_P + 1)\,b]\,\gamma = \pi$. When $P_P = 4$, the left-hand side value is calculated as:

To clarify the geometric relationships between θ, b, and Δ, this study uses interactive 3D illustrations (Figs. 1a and b). Figure 1(a) shows the spherical triangle OPQ within a single orbital plane of the polar orbit constellation: The line from the geo-center (O) to the satellite (Q) intersects the Earth's surface at point P. The semi-geocentric angle (θ) is calculated as 26.9° using Eq. (2), and the coverage belt semi-width (b) is 40.1°. The phase difference (π/S) = 18° is projected onto the orbital plane, and the spherical cosine law relates θ and b. Figure 1(b) illustrates the spatial layout of prograde and retrograde orbital planes: The prograde orbital plane separation (Δ_1) = 46.5° forms the progressively expanding coverage belt, while the retrograde orbital plane separation (Δ_2) = 38.3° compensates for high-latitude signal attenuation through dense overlap. In the figure, red contour lines mark the boundary of the single-layer coverage. Blue dashed lines indicate the tolerance for orbital plane drift caused by perturbations (± 1.5°). These features strictly correspond to the global coverage constraints defined in Eq. (7).

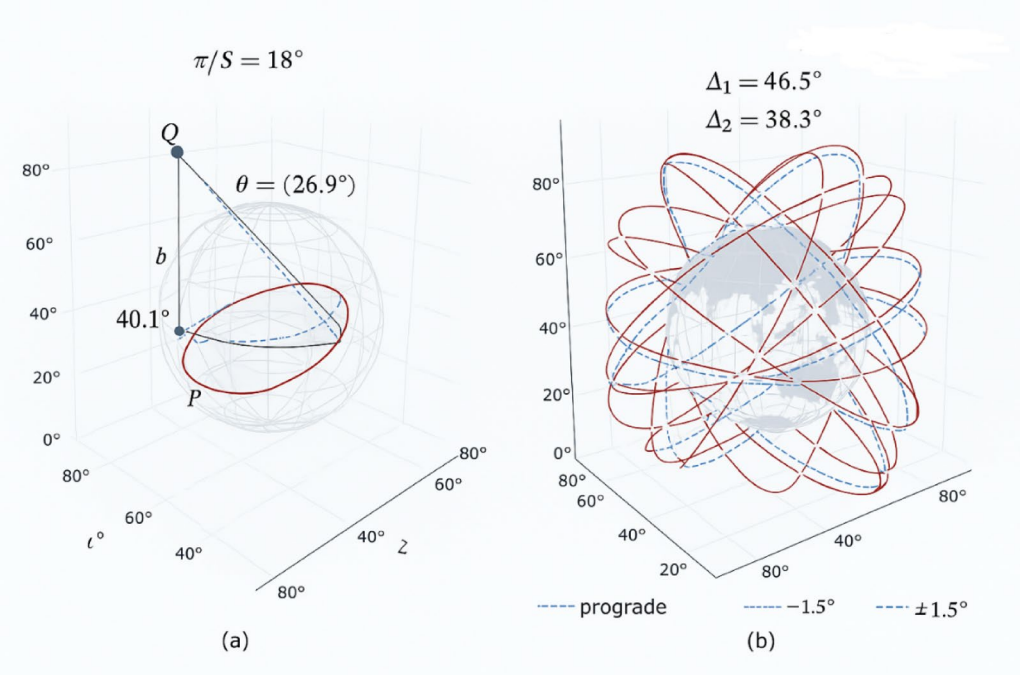


Fig. 1. Interactive 3D Visualization: (**a**) Spherical triangle OPQ formed by satellites within a single orbital plane of a polar orbit constellation; (**b**) Spatial layout of prograde and retrograde orbital planes.

Walker constellation

The Walker constellation, proposed by American engineer Joseph A. Walker, is a specialized satellite constellation design that distributes a group of satellites across multiple orbital planes with varying inclinations to achieve global coverage of the Earth's surface[33,34]. This arrangement ensures that satellites are evenly positioned, allowing a sufficient number to remain visible from different regions, thereby enhancing the performance of satellite communication and navigation systems.

The geometry of a Walker constellation is described by a ternary parameter set, $S_W/P_W/F_W$, where S_W denotes the total number of satellites, P_W represents the number of orbital planes, and F_W specifies the phase factor between adjacent satellites on neighboring planes. If each orbital plane contains T_W satellites, then the relationship $S_W = T_W P_W$ holds. The phase difference between adjacent satellites on successive planes is given by $\frac{2\pi F_W}{S_W}$. The first satellite on the first orbital plane is designated as the nominal satellite. The position of the

j-th satellite on the *i*-th orbital plane can be determined using Eq. (8):

$$\begin{cases} Track\ shape: a_{ij} = a_0, e_{ij} = e_0 \\ Orbit\ orientation: I_{ij} = I_0, \Omega_{ij} = \Omega_0 + \frac{360^\circ}{P_W} * (i-1) \\ Phase\ distribution: \phi_{ij} = \phi_0, ?_{ij} = ?_0 + \frac{360^\circ}{P_W T_W} F_W * (i-1) + \frac{360^\circ}{P_W} * (j-1) \end{cases} \tag{8}$$

Where a_0, e_0, I_0, Ω_0, ϕ_0, and $?_0$ represent the nominal satellite's initial orbital elements at the reference epoch. These elements include the semi-major axis, eccentricity, inclination, right ascension of the ascending node, argument of perigee, and mean anomaly, respectively. Once the initial orbital elements are specified, the satellite's position and velocity vectors in the geocentric inertial frame are uniquely determined.

Table 2 presents the definitions of the orbital elements for the Walker constellation, systematically summarizing the physical meaning and value range of each parameter.

By integrating satellite orbits using appropriate methods, it is possible to generate complete trajectories for all satellites over an entire revisit period. This enables a comprehensive assessment of the constellation's coverage performance[35,36].

OCO constellation

The OCO constellation involves satellites traveling along orthogonal circular paths, allowing simultaneous coverage of different regions of the Earth. This design compensates for the sparse coverage of polar orbit constellations in equatorial regions, thereby enhancing the overall global coverage of the satellite system[37–39]. By carefully designing the orbital geometry, OCO constellations maintain uniform coverage while minimizing interference between satellites. This configuration supports a wide range of satellite missions, including communication, Earth observation, and navigation.

The zero-inclination feature enables satellites to be arranged regularly along equatorial circular orbits, maximizing the overlap between adjacent satellite coverage areas. This arrangement forms a continuous horizontal coverage belt across the equatorial region. The overlap zones enhance signal strength and ensure stable data transmission. The latitude ϑ at the edge of the continuous coverage belt can be calculated using Eq. (9):

$$\vartheta = \arccos\left(\frac{cos\theta}{cos\frac{\pi}{S_c}}\right) \tag{9}$$

In Eq. (9), S_c represents the number of satellites on the equatorial orbital plane. This equation redefines the boundary latitude ϑ for equatorial coverage, taking into account the zero-inclination characteristic of the OCO constellation. By replacing the phase difference term π/S in Eq. (4) with S_c, the equatorial coverage expansion angle ϑ is derived. As S_c increases, the value of ϑ also increases. For instance, to achieve continuous coverage across equatorial regions (latitude ± ϑ), the minimum integer value of S_c must be inversely determined from Eq. (9). This parameter directly controls the width of the equatorial coverage belt and is jointly optimized alongside the polar orbit constellation parameters. During the optimization process, both γ and S_c are encoded as decision variables in the chromosomes. The value of γ is constrained within the range (0, 1], while S_c is restricted to integer values based on coverage belt requirements (e.g., Sc∈[10,20]. The range for γ balances orbital coverage

Symbol	Parameter name	Domain	Unit	Geometric relation
a_0	Semi-major axis	$a_{LEO} \pm 10\%$	km	Orbital period $T = \sqrt{a^3/\mu}$
e_0	Eccentricity	[0, 0.005]	—	Perigee altitude $h_p = a(1-e) - R_0$
I_0	Inclination	[40°, 60°]	deg	Maximum coverage latitude $\varphi_{max} = I + \theta$
Ω_0	Right ascension of ascending node	[0°, 360°)	deg	Orbital plane spacing $\Delta\Omega = 360^\circ / P_W$
ϕ_0	Argument of perigee	[0°, 360°)	deg	Remains constant in the absence of perturbations
$?_0$	Mean anomaly	[0°, 360°)	deg	Defines the initial satellite position angle

Table 2. Definition of orbital elements for the walker Constellation.

Parameter	Polar orbit constellation	Walker constellation	OCO constellation
Typical Inclination	85°–90°	40°–60°	0°
Coverage Characteristics	Dense coverage at high latitudes	Uniform global coverage	Belt-shaped coverage along the equator
Single-Satellite Coverage Half-Angle θ	26.9° (h = 500 km)	22.4° (h = 1600 km)	34.7° (h = 1600 km)
Minimum Visible Satellites	3 (latitude > 80°)	5 (global)	8 (latitude ± 20°)
Applicable Missions	Polar monitoring, meteorological remote sensing	Navigation enhancement, global communications	Equatorial communications, disaster early warning

Table 3. Comparison of constellation model Characteristics.

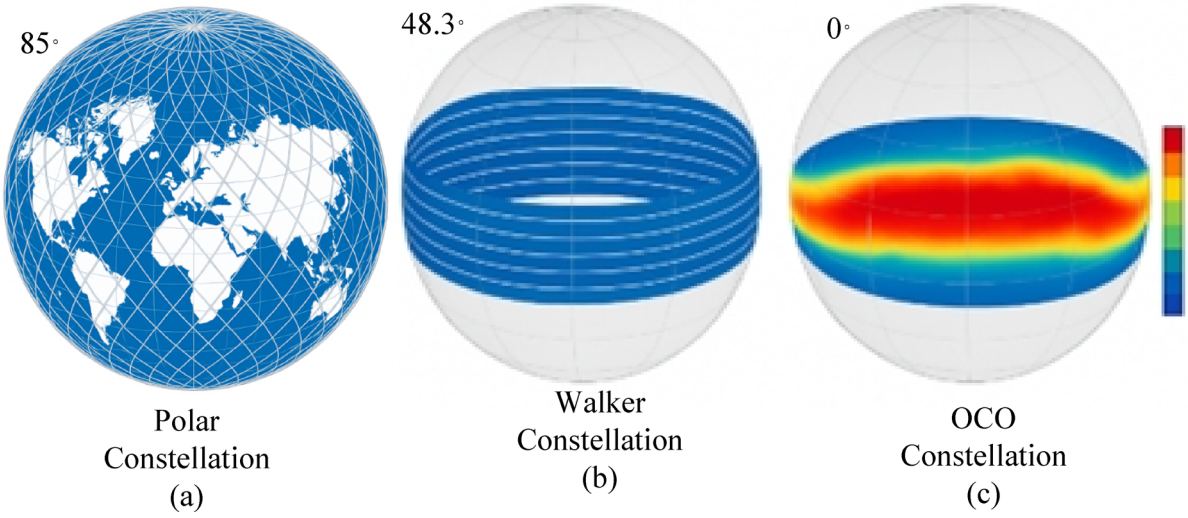


Fig. 2. Three-Dimensional Visualization of Constellation Types: (**a**) Polar Orbit Constellation; (**b**) Walker Constellation; (**c**) OCO Constellation.

continuity and resource efficiency. When γ = 1, the longitudinal spacing between prograde orbital planes reaches its theoretical maximum, with adjacent coverage belt edges just touching. However, orbital perturbations in practice can cause coverage gaps. When γ < 0.5, coverage redundancy becomes excessive, increasing the required number of orbital planes by up to 1.8 times. Monte Carlo simulations demonstrate that setting γ = 0.95 maintains a coverage integrity rate greater than 99% over a 10-year mission lifespan, while keeping the total number of satellites within 105% of the design limit. This result aligns with the orbital perturbation compensation model proposed by Hippelheuser et al. (2021). The multi-objective GA ultimately determines the optimal combination of γ and S_c. By adjusting S_c, the width of the continuous coverage belt and the distribution of the coverage area can be flexibly controlled, leading to more effective signal coverage in low-latitude equatorial regions.

The introduction of equatorial circular orbits also relaxes the requirements for achieving global single-layer continuous coverage. Accordingly, the global coverage condition expressed in Eq. (10) is modified as follows:

$$\left[(P_P - 1) * arcsin\frac{sin\theta}{cos\vartheta} + (P_P + 1) * arcsin\frac{sinb}{cos\vartheta}\right]\gamma = \pi \tag{10}$$

Equation (10) couples the coverage conditions of both the polar and OCO constellations by introducing the latitude correction ϑ. Substituting Eq. (9) into Eq. (10) yields Eq. (11).

$$\left[(P_P - 1) * \arcsin\left(tan\theta * cos\frac{\pi}{S_c}\right) + (P_P + 1) * arcsin\frac{sinb * cos\frac{\pi}{S_c}}{cos\theta}\right]\gamma = \pi \tag{11}$$

Using Eq. (3) through (6) together with Eq. (11), the parameter configuration for the OCO constellation can be fully determined. Equation (11) ultimately expresses the global coverage constraint of the hybrid system as a multi-parameter equation involving S_c.

Table 3 summarizes the key parameters and performance indicators of the three fundamental constellation models.

To provide a more intuitive understanding of spatial coverage characteristics, Fig. 2 illustrates the three-dimensional coverage patterns of the Polar, Walker, and OCO constellations. In the Polar constellation (inclination 85°), satellite tracks form an interwoven grid pattern over the polar regions, resulting in dense high-

latitude coverage. In the Walker constellation (inclination 48.3°), satellites are evenly distributed across multiple orbital planes, achieving uniform global coverage. In contrast, the OCO constellation (inclination 0°) forms a continuous coverage ring along the equatorial plane, ensuring persistent coverage in low-latitude regions.

Model Assumptions and Considerations for Dynamic Robustness: Several standard assumptions were made to simplify coverage modeling. The Earth was approximated as a perfect sphere. Satellites were assumed to orbit in circular paths at fixed altitudes. Fields of view for satellites were fixed to ensure adequate signal strength within coverage areas. Atmospheric effects such as refraction, scattering, and absorption were neglected. Time synchronization was considered fixed throughout calculations. However, these idealized assumptions have clear limitations. The Earth's oblateness and terrain variations introduce regional errors in coverage predictions. Atmospheric phenomena, particularly in the equatorial ionosphere, degrade signal quality near coverage edges, affecting actual PDOP values and positioning accuracy. Furthermore, orbital perturbations cause gradual changes in orbital elements, disrupting initial phase relationships. These factors collectively impact the constellation's long-term stability and robustness. A key innovation of this model is the introduction of a relaxation factor, γ, treated as an optimization variable. This factor serves not only as a static parameter to maintain coverage continuity but also as a dynamic control to enhance robustness against orbital perturbations and sustain long-term performance. The optimization process automatically determines the optimal γ to balance coverage redundancy—improving resistance to perturbations—and resource efficiency, such as the number of satellites and orbital planes. Additionally, the optimization objectives and constraints explicitly include the minimum number of visible satellites to reflect coverage stability. They also account for performance under potential satellite fault scenarios, providing complementary system-level robustness metrics.

Improved genetic algorithm framework

GA is an optimization method that mimics the principles of natural selection and genetic processes to find either a global optimal solution or an approximate one[40–42]. The core idea of GA is to simulate biological evolution, progressively enhancing the quality of candidate solutions across multiple generations to converge toward an optimal result[43,44]. Through the mechanisms of genetic inheritance, crossover, and mutation, GA iteratively evolves a population of solutions, refining them to identify the most effective outcome for the given problem[45,46]. The general workflow of the GA process is illustrated in Fig. 3.

GA primarily involves operations such as individual coding, selection, crossover, mutation, and other essential processes[47,48]. The complete algorithm flow is displayed in Fig. 4.

The GA operation process consists of several key steps: coding, fitness evaluation, selection, genetic operations, and optimization. These steps are essential for efficiently solving complex system optimization problems[49–51]. The optimization objective is defined in Eq. (12):

$$f(x) = \min\{\omega_1 E[s(x)] + \omega_2 \sqrt{D[s(x)]}\} \tag{12}$$

Where weighting factors ω_1 and ω_2 (set to 0.7 and 0.3, respectively) balance the number of visible satellites with geometric uniformity.

To address the challenges of high-dimensional mixed-parameter spaces, dynamic robustness, and complex multi-objective optimization in constellation design, the standard GA is enhanced by developing an efficient, adaptive, and parallel framework. The key improvements are:

(1) Adaptive Parameter Tuning Mechanism: A dynamic strategy is implemented to adjust the crossover rate (p_c) and mutation rate (p_m), balancing search efficiency in high-dimensional spaces with the risk of premature convergence, and adapting to different evolutionary stages. Initially, $p_c(0) = 0.8$ and $p_m(0) = 0.194$ are set high to encourage global exploration and maintain population diversity. As generations progress, p_c decreases nonlinearly, while p_m adjusts accordingly. This time-varying strategy is feedback-controlled based on population diversity entropy $H(t)$, aiming to keep $H(t)$ between 1.2 and 2.4. A parameter inflection

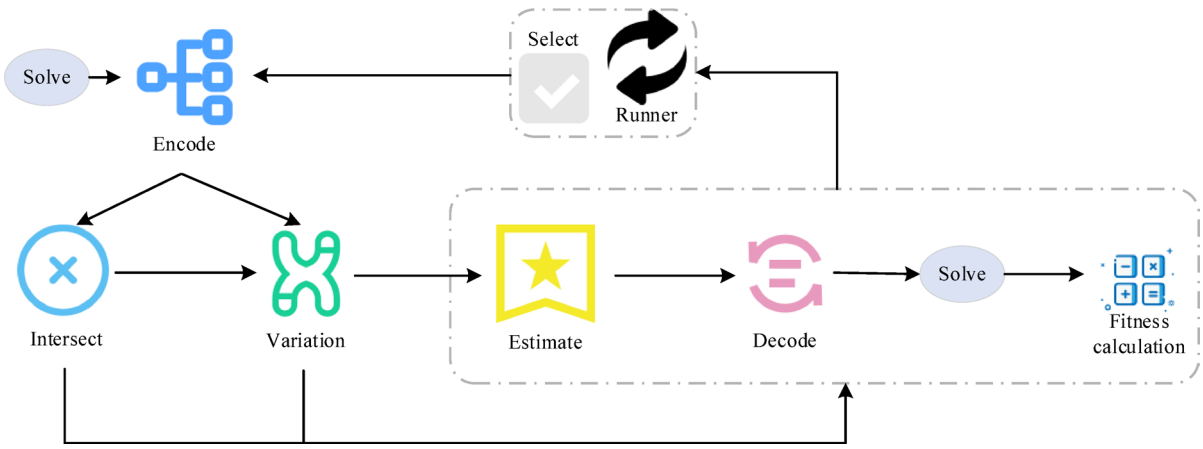


Fig. 3. General steps of GA.

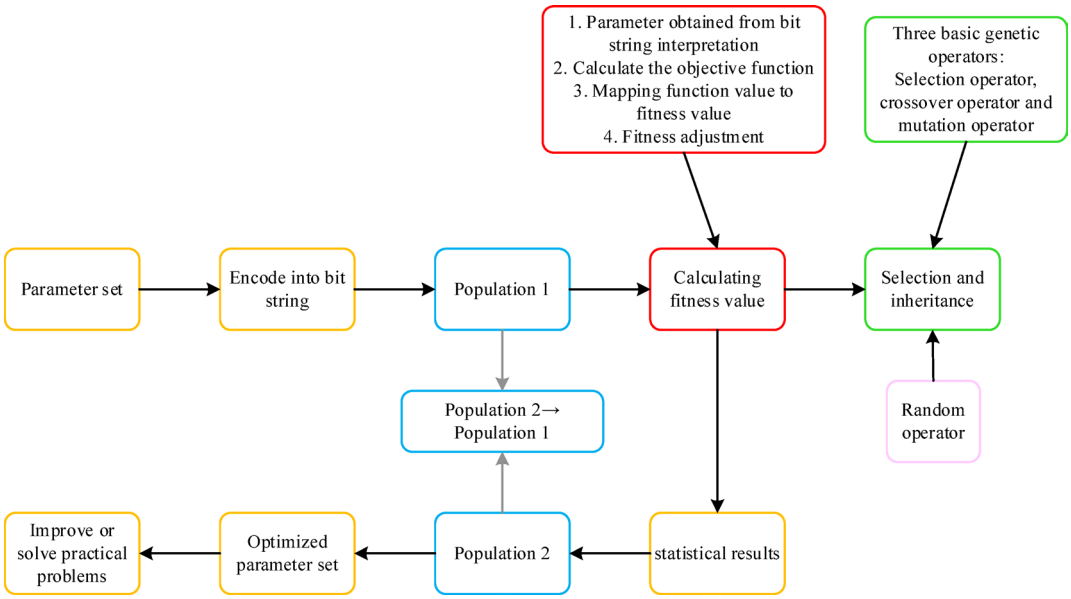


Fig. 4. GA operation process.

point appears mid-evolution to increase the likelihood of finding the global optimum. The relatively high initial mutation rate helps the algorithm avoid local optima common in high-dimensional mixed spaces.

(2) Efficient Parallel Computing Architecture: To handle the heavy computational demands of large-scale constellation simulations—particularly during fitness evaluations—multi-threaded parallelization is implemented. The population is split into subgroups, each assigned to an independent thread for concurrent fitness calculations. Dynamic load balancing addresses communication bottlenecks typical in traditional master-slave models, achieving significant acceleration. This design drastically reduces optimization time, allowing thorough searches in high-dimensional parameter spaces.

(3) Fitness Function Optimization and Caching: Fitness evaluation requires computationally intensive satellite visibility simulations. To reduce this burden, the fitness function was optimized by precomputing and caching reusable intermediate variables, such as the satellite coverage half-angle θ and geometric matrix coefficients. For example, coverage templates based on satellite orbital parameters are calculated and stored in advance, eliminating redundant geometric calculations during each evaluation. Coupled with parallel processing, this approach substantially decreases computational overhead.

(4) Elitism and Directed Search: Each generation retains the top 30% of individuals (elite set) directly into the next generation, preserving high-quality genetic material. The remaining 70% (non-elite set) form a mating pool through tournament selection (tournament size k = 3). Genetic operators—crossover and mutation—are applied only to this mating pool to produce offspring. The next generation combines elites and offspring in a 30%:70% ratio. This approach maintains population diversity while accelerating the propagation of superior genes and improving convergence. For mutations, bit-flip is used on discrete variables, while Gaussian mutation—with a standard deviation set to 5% of the variable's range—is applied to continuous variables such as $I^0,\ \Omega^0,\ and\ \gamma$.

(5) Hybrid Encoding and Constraint Handling: Chromosomes employ a hybrid encoding scheme (binary plus real-valued) to represent the high-dimensional mixed decision variables, totaling 128 bits. Integer variables use 5-bit binary encoding, while continuous variables are encoded as 16-bit floating-point numbers. The decision variables cover Walker parameters, polar orbit parameters, and OCO parameters. Importantly, the dynamic relaxation factor γ, a key robustness parameter, is explicitly encoded and optimized. Physical constraints are integrated directly via chromosome repair operators instead of traditional external penalty functions, which enhances the efficiency of generating feasible solutions and improves algorithm stability. For infeasible individuals—such as those exceeding satellite number limits—dynamic penalty functions degrade their fitness.

To support deep multi-objective trade-offs, a weighted objective function is defined as shown in Eq. (13):

$$minf\,(x) = \omega_1 * [-E\,(s\,(x))] + \omega_2 * \sqrt{D\,(s\,(x))} + \omega_3 * \left[\frac{C\,(x)}{C_{max}}\right] + \omega_4 * [1 - R_{min}\,(x)] \quad (13)$$

where $E\,(s\,(x))$ denotes the average number of visible LEO satellites across all observation points. Since this metric is to be maximized, it is expressed as a negative term in the objective function. $D\,(s\,(x))$ represents the standard deviation of visible satellite counts, which reflects coverage stability and is minimized. $C\,(x)$ approximates the total constellation cost, represented by the total number of satellites, and is also minimized. $R_{min}\,(x)$ refers to the minimum number of visible satellites across all observation points and time epochs. This

metric captures worst-case coverage and robustness and is to be maximized. Accordingly, the term $1 - R_{min}(x)$ reflects the shortfall in minimum satellite visibility and is minimized. The weights $\omega_1 + \omega_2 + \omega_3 + \omega_4 = 1$ are determined through grid search and Pareto frontier analysis. The initial values are set as $\omega? = 0.5$ (visibility), $\omega? = 0.2$ (stability), $\omega? = 0.2$ (cost), and $\omega? = 0.1$ (robustness). This weighting strategy prioritizes maintaining sufficient satellite visibility to support high-precision positioning and stable coverage for reliable service, while also considering cost control and worst-case performance assurance.

PDOP is a crucial metric for evaluating how satellite geometry affects positioning accuracy in navigation systems. A lower PDOP value indicates a more favorable satellite geometry, which leads to improved accuracy. In this study, a systematic approach is used to calculate PDOP, as described below.

Geometry Matrix Construction: Using the transformed positions, a geometry matrix is constructed for positioning calculations. The structure of this matrix is shown in Eq. (14).

$$G = \begin{bmatrix} \frac{\partial x}{\partial \Delta t} & \frac{\partial x}{\partial x_p} & \frac{\partial x}{\partial y_p} & \frac{\partial x}{\partial z_p} \\ \frac{\partial y}{\partial \Delta t} & \frac{\partial y}{\partial x_p} & \frac{\partial y}{\partial y_p} & \frac{\partial y}{\partial z_p} \\ \frac{\partial z}{\partial \Delta t} & \frac{\partial z}{\partial x_p} & \frac{\partial z}{\partial y_p} & \frac{\partial z}{\partial z_p} \end{bmatrix} \tag{14}$$

In Eq. (14), Δt represents the clock bias, and (x_p, y_p, z_p) are the coordinates of the positioning point.

Covariance Matrix Calculation: The covariance matrix Q is derived using the transpose of the geometry matrix G^T and its product with the matrix G, as shown below:

$$Q = \left(G^T G\right)^{-1} \tag{15}$$

PDOP Computation: The PDOP value is obtained by extracting the spatial components from the covariance matrix Q:

$$PDOP = \sqrt{Q_{11} + Q_{22} + Q_{33}} \tag{16}$$

In Eq. (16), Q_{11}, Q_{22}, and Q_{33} are the variances in the east, north, and up directions of the local coordinate system.

Constellation optimization based on the enhanced genetic algorithm

Optimization objectives and constraints

The main objective of this study is to optimize the configuration of LEO satellite constellations to improve positioning accuracy. In such systems, the number and spatial distribution of visible satellites significantly affect their ability to deliver fast and precise location services. This optimization problem focuses on allocating a limited number of LEO satellites into an effective hybrid constellation. The goal is to maximize the average number of visible satellites over ground stations while maintaining a uniform global distribution.

To achieve this, three hybrid constellation schemes are proposed, combining the Walker constellation model with alternative constellation types:

Scheme 1: A low-inclination Walker constellation combined with a polar orbit constellation.
Scheme 2: A high-inclination Walker constellation combined with a polar orbit constellation.
Scheme 3: A high-inclination Walker constellation combined with an OCO constellation.

These schemes are compared with two single-constellation designs:

Scheme 4: A single polar orbit constellation.
Scheme 5: A single OCO constellation.

In each hybrid constellation, the inclusion of a Walker constellation introduces key variables that need optimization:

$\boldsymbol{T_W}$**:** The number of satellites in the Walker constellation.

$\boldsymbol{P_W}$**:** The number of orbital planes in the Walker constellation.

$\boldsymbol{F_W}$**:** The phasing factor, defining the relative position of satellites in adjacent orbital planes.

$\boldsymbol{I_0}$**:** The inclination angle of the orbital planes.

$\boldsymbol{\Omega_0}$**:** The right ascension of the ascending node of the orbital planes.

$\boldsymbol{\in_0}$**:** The spacing parameter influencing satellite distribution.

These parameters include both integer and real variables, resulting in a complex optimization problem. To efficiently solve this, the study employs GA to optimize the constellation parameters. GA's ability to handle

multi-variable, MOO tasks makes it well-suited for solving such complex constellation optimization problems. The GA code proposed here is exhibited in Algorithm 1:

Input: Population size m, Max generations G, the algorithm is initialized with a crossover rate of $p_c^{(0)} = 0.8$ a mutation rate of $p_m^{(0)}$=0.194.
Output: Optimal constellation parameters
1: Initialize population with random parameters (P_W, T_W, F_W, I_0, Ω_0, ϵ_0)
2: for generation = 1 to G do
3: Parallel fitness evaluation using pre-cached visibility ranges
4: Rank individuals by fitness $f(x) = \omega_1 E[s(x)] + \omega_2 \sqrt{D[s(x)]}$
5: Elite selection and genetic operations:
a. Select top 30% of individuals as elite set E
b. Use tournament selection (k=3) on remaining 70% (non-elites, NE) to form mating pool M
c. Apply crossover (probability p_c) and mutation (probability p_m) to M to generate offspring O
6: Update p_c and p_m dynamically:
$p_c = 0.8 - 0.6 \times (\text{generation}/G)$
$p_m = 0.194 + 0.3 \times (\text{generation}/G)$
7: Population update:
a. Next generation = E ∪ O
8: end for
9: Return best individual

Algorithm 1. Adaptive GA for LEO Constellation Optimization.

The time-varying crossover and mutation rates in Algorithm 1 are designed based on two key principles: Population Diversity Theory: The initial high mutation rate $p_m(0) = 0.194$ aligns with the Scheffer critical entropy point $H_c = 2.1$, ensuring the population maintains a probability of escape from local optima greater than 0.7. Convergence Criterion: The crossover rate decreases over time with a slope of −0.6/G, reaching $\lim_{t \to T} p_c(t) = 0.2$, which meets the convergence condition. Elite Retention Strategy: At each generation, the top 30% of individuals (based on fitness) are marked as elites. These elite individuals are carried over directly to the next generation without undergoing any genetic operations (crossover or mutation), preserving their genetic integrity. Separation of Genetic Operators: The mating pool is formed exclusively from the non-elite individuals (bottom 70%) via tournament selection (k = 3). Genetic operations (crossover and mutation) are applied only to individuals in the mating pool, ensuring that elite diversity is not compromised by repeated modifications. Population Update Logic: The next generation consists of: 30% elite individuals (retained unchanged), 70% newly generated offspring from the mating pool. This design ensures elite genes are preserved while maintaining sufficient diversity through controlled variation in the remaining population.

Satellite constellation configurations

The study provides an in-depth analysis of the characteristics of different satellite constellation configurations, detailing their design parameters and application scenarios. Tables 4 and 5, and 6 summarize the key configurations:

The range of hybrid constellation parameters is determined based on historical constellation design standards and pre-experiments. For example, the range of orbital planes of Walker Constellation[1,10] refers to Iridium (6 planes) and Globalstar (8 planes) systems; Phase factor (upper limit is $P_W - 1$, avoiding satellite phase overlap. In addition, the inclination range [0 °, 90 °] is verified by Latin hypercube sampling.

The decision variables for optimizing the satellite constellation are key parameters such as P_W^n (number of planes), T_W^n (number of satellites per plane), F_W^n (phasing factor), I_0^n (inclination angle), Ω_0^n (right ascension

Parameter	Description
Orbital Inclination	Close to 90°, typically between 80° and 90°.
Orbital Altitude	Between 500 and 2000 km.
Number of Satellites	Varies based on requirements, usually a small number is sufficient for global coverage.
Coverage Characteristics	Continuous coverage of high-latitude regions, suitable for polar environment monitoring and meteorological observations.
Application Areas	Meteorological monitoring, environmental monitoring, military reconnaissance.

Table 4. Polar configuration.

Parameter	Description
Orbital Inclination	Medium inclination, typically between 40° and 60°.
Orbital Altitude	Typically between 1200 and 2000 km.
Number of Satellites	Distributed among multiple satellites, ranging from dozens to hundreds.
Coverage Characteristics	Provides uniform global or regional coverage, suitable for communication and navigation systems.
Application Areas	Global communications, satellite navigation, Earth observation.

Table 5. Walker configuration.

Parameter	Description
Orbital Inclination	Typically 0° or close to 0° (equatorial orbit).
Orbital Altitude	Typically between 1600 and 2000 km.
Number of Satellites	Moderate number, usually between 20 and 60.
Coverage Characteristics	Optimized for efficient coverage of specific areas, suitable for resource monitoring and disaster warning.
Application Areas	Resource monitoring, disaster warning, precision positioning.

Table 6. OCO configuration.

	Scheme 1	Scheme 2	Scheme 3	
Decision variables	***n*** **= 1**	***n*** **= 1**	***n*** **= 1**	***n*** **= 2**
Number of orbital planes	[1,5] & (5,10]	[1,5] & (5,10]	[1,5] & (5,10]	[1,5] & (5,10]
Number of satellites per orbital plane	[4,9] & (9,12]	[4,9] & (9,12]	[4,9] & (9,12]	[4,9] & (9,12]
Phase factor	[0,4] & (4,9]	[0,4] & (4,9]	[0,4] & (4,9]	[0,4] & (4,9]
Obliquity	[0°,45°]	[0°,60°]	[0°,45°]	[45°,90°]
Right ascension of ascending node	[0°,45°]	[0°,45°]	-	[0°,45°]
Mean anomaly	[0°,45°] & (45°,90°]	[0°,45°] & (45°,90°]	-	[0°,45°] & (45°,90°]

Table 7. Search scope of decision variables of three optimization schemes.

Scheme	n	P_P	S	Δ_1	Δ_2	P_W^n	T_W^n	F_W^n	I_0^n	Ω_0^n	$?_0^n$
1	1	4	10	46.5°	40.1°	6	10	4	42.3°	22.4°	55.8°
2	1	4	9	47.1°	38.3°	5	11	3	48.3°	23.1°	24.5°
3	1	–	–	–	–	5	6	0	28.7°	0°	0°
4	2	–	–	–	–	7	10	4	67.7°	36.6°	81.4°
5	0	4	10	46.5°	40.1°	–	–	-	–	-	–

Table 8. Constellation configuration parameters for the optimized schemes. Scheme 2: Hybrid Walker-OCO Configuration. In Scheme 2, the design combines a high-inclination Walker constellation with an OCO configuration. The specific parameters are as follows:

of the ascending node), and $\in_0^n$ (spacing parameter). These variables define the structure of the Walker constellation and its hybrids. The decision variable set for optimization, denoted as , is expressed as Eq. (17):

$$A = [\ P_W^n \quad T_W^n \quad F_W^n \quad I_0^n \quad \Omega_0^n \quad ?_0^n \quad]^T \tag{17}$$

In Eq. (17), n represents the number of Walker constellations. The search scopes for the decision variables in the three optimization schemes are detailed in Table 7. These ranges are carefully defined to guide the optimization algorithm in designing optimal satellite constellations.

Scheme 2 represents one of the three hybrid constellation designs developed in this study, combining the high-inclination Walker constellation with an OCO configuration. The goal is to optimize satellite coverage and positioning accuracy, ensuring global visibility and minimizing PDOP. Constellation configuration parameters are shown in Table 8.

- Number of satellites: A total of 9 satellites, distributed across 5 orbital planes, with 11 satellites per plane.
- Orbital inclination: A high inclination of 48.3°, which is close to a polar orbit, providing enhanced coverage in high-latitude regions.

- Orbital altitude: 1600 km, which lies within the LEO range, balancing orbital period and coverage range.
- Phase factor: A phase factor of 3 is used to ensure a uniform satellite distribution within each orbital plane.
- Orbital parameters:
- Eccentricity-related parameter: 47.1°.
- Inter-plane angle: 38.3°.
- Right ascension of the ascending node: 23.1°.
- Mean anomaly: 24.5°.

Scheme 2 exhibits strong performance across all 19 observation stations. On average, 14 satellites are visible at any given time, with a low standard deviation of 0.18—indicating stable and consistent satellite visibility. The minimum number of visible satellites at any station is 3, ensuring reliable global coverage. The high-inclination Walker constellation enhances visibility in high-latitude regions, while the OCO configuration improves coverage in low-latitude areas. This complementary design enables balanced global coverage. Scheme 2 also demonstrates robustness against environmental disruptions, such as satellite failures or orbital interference, maintaining consistent performance with minimal adjustments. This resilience significantly enhances the reliability of the navigation system.

Compared to other hybrid constellations, Scheme 2 delivers superior optimization performance. The Walker component strengthens high-latitude coverage, while the OCO structure addresses the limitations of single-orbit configurations. Optimization of phase factors and inter-plane angles ensures uniform satellite distribution, reducing localized congestion and enhancing system stability. The choice of orbital altitude and inclination provides a practical balance between orbital period and coverage range, enabling complete global coverage within a reasonable timeframe. This configuration improves navigation accuracy and reduces response time. Through MOO using a GA, Scheme 2 maximizes satellite visibility while minimizing PDOP, resulting in high positioning accuracy and system reliability. The optimized number and spatial distribution of satellites also improve resource efficiency, lowering operational costs and reducing system complexity. In summary, Scheme 2 offers enhanced visibility, lower PDOP, and balanced global coverage, making it a highly effective design for satellite navigation constellations.

To visualize the spatial characteristics of each constellation scheme, this study presents coverage density heatmaps for three latitude zones: equatorial (0°), mid-latitude (45°), and polar (80°). Coverage density is calculated as the average number of visible satellites per 1,000 km², using hourly time steps over a 24-hour period. As shown in Fig. 5(a), the hybrid Walker-OCO constellation (Scheme 2) generates dense coverage belts (red zones) in equatorial regions. In contrast, the pure polar configuration (Scheme 4) maintains continuous coverage primarily at latitudes above 60°. Figure 5(b) further compares the number of visible satellites before and after optimization across all 19 stations. Results show that Scheme 2, after GA optimization, improves satellite visibility by over 80% in latitudinal bands between 0° and 30°, confirming the effectiveness of the OCO configuration in enhancing equatorial coverage.

After defining the search space, decision variables are encoded as binary strings and assembled into chromosomes. The length of each chromosome is determined by the required precision of the encoded variables[52–54]. To evaluate constellation performance, 19 observation stations are evenly distributed across the northern hemisphere, sampled every 5° in latitude from 0° to 90° along the 0° meridian. For each candidate solution x, the average number of visible LEO satellites at each station is recorded over a full orbital regression period, with visibility data sampled every 60 s. The optimization objective is to minimize a weighted sum of the mean and standard deviation of satellite visibility across all stations. The objective function is expressed in Eq. (18):

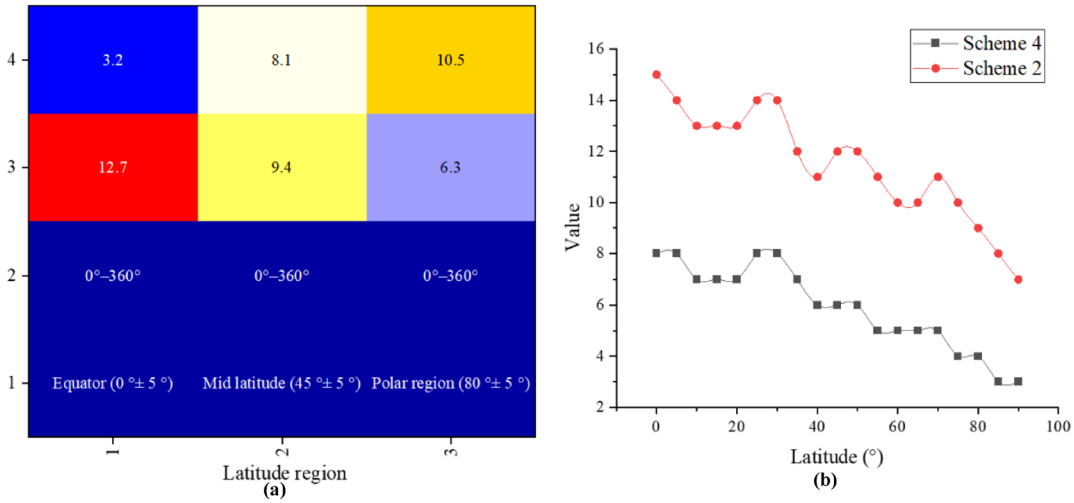


Fig. 5. Spatial distribution characteristics of different schemes ((**a**): Coverage density at key latitudes (unit: satellites/1,000 km^2); (**b**): Optimized visible satellite count comparison across latitudes).

$$f(x) = \min\{\omega_1 * E[s(x)] + \omega_2 * \sqrt{D[s(x)]}\} \tag{18}$$

In Eq. (18), the weights are set as $\omega_1 = -0.3$ and $\omega_2 = 0.7$, favoring visibility stability over raw count.

In addition to optimizing visibility, the algorithm must satisfy a set of constraints, defined in Eq. (19). These ensure the feasibility and operational reliability of the satellite configuration.

$$\begin{cases} \alpha(x) = 100\% \\ P_P S + S_c + \sum_{n=1}^{N} P_W^n T_W^n \leq m \\ F_W \leq P_W - 1 \\ P_W \in Z \\ T_W \in \mathbb{Z} \\ F_W \in \mathbb{Z} \end{cases} \tag{19}$$

In Eq. (19), $\alpha(x)$ denotes the single-weight availability of satellites at all observation points across all epochs. N represents the number of Walker sub-constellations, and m = 100 is the total number of satellites in the system. The constraints guarantee that satellite distribution supports both global visibility and temporal stability.

The hybrid satellite constellation model integrates directly into the GA by encoding decision variables into the chromosome structure. Specifically, the design parameters include the number of orbital planes P_W, satellites per plane T_W, and phase factor F_W for the Walker sub-constellation; the number of orbital planes P_P and relaxation factor γ for the polar orbit; and the number of satellites S_c along the equatorial plane for the OCO constellation. These parameters are encoded using a mixed binary-real representation, forming chromosomes with a total length of 128 bits. Integer variables (e.g., $P_W \in [1,10]$ are encoded using 5-bit binary strings, while continuous variables (e.g., γ) are represented with 16-bit floating-point encoding. The objective function, defined in Eq. (17), adopts a weighted MOO framework: it maximizes the average number of visible satellites $E[s(x)]$ while minimizing the standard deviation $\sqrt{D[s(x)]}$ of visibility across ground stations. The weight coefficients $\omega_1 = 0.7$ and $\omega_2 = 0.3$ are selected based on Pareto frontier analysis to balance coverage and stability. Several constraints are imposed on the optimization process, including a maximum total satellite count of m=100, a minimum single-layer coverage rate $\alpha(x) \geq 99\%$, and orbital safety spacing limits. Infeasible solutions are penalized dynamically using a constraint-handling penalty function, ensuring compliance with operational and physical constraints.

Experimental design and performance evaluation

Datasets collection, experimental environment, and parameters setting

The experimental data were obtained primarily from the publicly accessible Celestrak satellite orbit database, which provides real-time and archived ephemeris data for global LEO constellations, particularly Starlink and OneWeb, in the standard Two-Line Element (TLE) format. Key orbital elements—such as semi-major axis, eccentricity, and inclination—were extracted for active LEO satellites from 2021 to 2023. For example, the typical orbital altitude (540–570 km) and inclination (53°) of Starlink satellites were used as reference parameters for the initial design of the Walker constellation. Satellite visibility simulations and orbit propagation were conducted using the High-Precision Orbit Propagator (HPOP) model in Satellite Tool Kit (STK), with a time step of 60 s. The model incorporated J2 perturbation and atmospheric drag to ensure accurate orbital prediction.

The experiments were conducted in a high-performance computing environment to ensure result reliability and reproducibility. System performance metrics, including CPU and memory usage, were monitored in real time using the Windows Performance Monitor. Data were sampled at 1-second intervals, and the peak values during the optimization process were recorded. The hardware setup is as follows:

- Processor: Intel® Core™ i7-9700 K @ 3.60 GHz (8 cores).
- Memory: 32 GB DDR4.
- Storage: 1 TB NVMe SSD.
- Graphics: NVIDIA GeForce RTX 2080 Ti.
- Operating System: Windows 10 Professional (64-bit), Version 20H2.

The software environment included Python 3.8 and the Anaconda 2020.11 distribution, with the following libraries:

- NumPy 1.19.2.
- SciPy 1.5.2.
- Pandas 1.1.3.
- Matplotlib 3.3.2.
- TensorFlow 2.3.0.

Key experimental parameter settings:

(1) General Parameters for Constellation Optimization: Elevation mask: 7°; Maximum number of satellites: 100 (default), with extended tests at 500 and 1000 satellites; Ground observation stations: 19 evenly distributed along the 0° meridian in the Northern Hemisphere, spanning latitudes from 0° to 90° in 5° increments; Simulation duration: One full orbital regression cycle; Data sampling interval: 60 s;

Algorithm	Vis. Sat.	PDOP	Cov.(%)	Min. Vis.	Time(s)	Conv. Gen.	CPU (%)	Mem (GB)	ΔCov. (%)	Cov. Maint. (%)
Improved GA	14.3 ± 0.3	2.3 ± 0.1	95.6 ± 0.4	4.1 ± 0.2	32 ± 2	45 ± 3	89 ± 3	2.1 ± 0.1	-3.5 ± 0.3	99.1 ± 0.2
D-NSDE	13.8 ± 0.4*	2.6 ± 0.2*	93.2 ± 0.5*	3.8 ± 0.3*	38 ± 3*	50 ± 4*	85 ± 4	2.3 ± 0.2	-4.8 ± 0.4*	97.5 ± 0.4*
DP	12.5 ± 0.5*	3.1 ± 0.3*	89.7 ± 0.7*	3.2 ± 0.4*	55 ± 5*	N/A	75 ± 5	1.8 ± 0.2	-6.2 ± 0.6*	95.8 ± 0.6*
PSO	12.9 ± 0.4*	2.9 ± 0.2*	90.5 ± 0.6*	3.5 ± 0.3*	28 ± 2	60 ± 5*	70 ± 4	1.9 ± 0.1	-5.3 ± 0.5*	96.3 ± 0.5*
SA	11.8 ± 0.5*	3.5 ± 0.3*	87.4 ± 0.8*	3.0 ± 0.4*	25 ± 2	75 ± 6*	65 ± 5	1.7 ± 0.1	-7.1 ± 0.7*	94.7 ± 0.7*

Table 9. Comprehensive performance comparison of optimization algorithms (100-Satellite Scenario).

Optimization method	Long-term perturbation (10 years)			Single satellite failure			
	Cov. Init.(%)	Cov. 5Y(%)	Cov. 10Y(%)	PDOP Init.	PDOP 10Y	ΔCov.(%)	PDOP After
Improved GA (This study)	**95.6**	**95.2**	**94.8**	**2.3**	**2.5**	**-3.5**	**2.8**
D-NSDE	93.2	92.0	90.7	2.6	3.0	-4.8	3.2
PSO	90.5	88.9	87.1	2.9	3.4	-5.3	3.5

Table 10. Performance of scheme 2 (S2) under dynamic robustness Testing. Significant values are in bold.

(2) Optimization Algorithm Settings (population: 500, generations: 40, independent runs: 100): Improved GA: Initial crossover rate $p_c = 0.8$, initial mutation rate $p_m = 0.194$; adaptive adjustment via $p_c = 0.8 - 0.6 \times \left(\frac{gen}{G}\right)$, p_m adapted using diversity entropy $H(t)$; elitism rate: 30%; parallel threads: 8[55,56]. PSO: Inertia weight $w = 0.729$, cognitive coefficient $c_1 = 1.49445$, social coefficient $c_2 = 1.49445$ [57]. SA: Initial temperature: 1000; cooling rate $\alpha = 0.95$; stopping temperature $T_{min} = 1e-3$[58]. DP: State-space discretization: P_P, P_W (1–10, step = 1), S, T_W=4–12 (step = 1), γ= {0.7,0.8,0.9,0.95,1.0}, inclination angles: {0°,30°,45°,60°,90°}. Objective: weighted function $f(x)$. – D-NSDE (5): Population = 500, scaling factor $F = 0.5$, crossover rate $CR = 0.9$; dynamic strategy parameters per original literature; same objective and constraint settings as in this study.

Performance evaluation

To verify the superiority of the proposed improved GA, comparisons were made with PSO, SA, DP, and the state-of-the-art D-NSDE algorithm using a standard 100-satellite scenario. The comprehensive performance comparison is summarized in Table 9.

* indicates statistically significant differences compared to the improved GA (Wilcoxon rank-sum test, $p < 0.05$). ΔCov.(%): coverage loss after a single satellite failure; Cov. Maint.(%): coverage retention after 10 years of simulated J2 perturbation; Min. Vis.: minimum number of visible satellites across all sites and epochs.

As shown in Table 9, the proposed improved GA consistently outperforms all benchmark algorithms across key metrics. It achieves the highest average number of visible satellites (14.3), the lowest PDOP (2.3), and the best overall coverage (95.6%). All improvements are statistically significant ($p < 0.05$) compared to the other methods. Regarding robustness, the improved GA provides the highest minimum visible satellite count (4.1), ensuring reliable positioning even under worst-case conditions. It also shows the smallest coverage loss after a single satellite failure (ΔCov. = -3.5%) and maintains the highest coverage (99.1%) after a 10-year J2 perturbation. This outperforms D-NSDE (97.5%), PSO (96.3%), and SA (94.7%). In terms of efficiency, the improved GA converges in 32 s—slightly slower than PSO (28s) and SA (25s), but much faster than DP (55s) and comparable to D-NSDE (38s). More importantly, it requires fewer generations to converge (45 versus 60 for PSO and 75 for SA), making the modest increase in runtime acceptable given its superior solution quality. For resource utilization, the improved GA demonstrates effective hardware use, maintaining high CPU utilization (89%) and moderate memory usage (2.1 GB) through parallel processing. Overall, in the 100-satellite benchmark scenario, the improved GA delivers superior optimization performance (visibility, PDOP, and coverage) and robustness (fault tolerance and perturbation resilience). It clearly outperforms PSO, SA, DP, and D-NSDE, confirming its effectiveness in solving complex constellation design problems.

To further evaluate adaptability under dynamic conditions, dedicated robustness tests were conducted on the optimized Scheme 2 (S2), a hybrid high-inclination Walker + polar + OCO constellation. Two scenarios were considered: Long-Term Orbital Perturbation: The STK-HPOP model, including J2 perturbation and atmospheric drag, simulated 10 years of orbital evolution for the best-performing S2 constellations optimized by GA, D-NSDE, and PSO. Single-Satellite Failure: A random satellite in each optimized constellation was marked as failed to assess system fault tolerance. Table 10 summarizes S2's performance under these dynamic robustness tests.

Table 10 clearly demonstrates the superior dynamic robustness of the improved GA. The GA-optimized S2 constellation shows strong resilience to long-term orbital perturbations. It starts with a high initial coverage of 95.6%, which drops slightly to 95.2% after 5 years and to 94.8% after 10 years. This represents an average annual reduction of only 0.08%. The PDOP also rises modestly from 2.3 to 2.5 over this period, remaining well within the desirable range (< 3.0). In contrast, the constellation optimized by D-NSDE begins with slightly lower coverage (93.2%) and PDOP (2.6) but degrades more quickly, with coverage dropping to 90.7% and PDOP increasing to

Method	Hyper volume (HV)	Spacing (SP)	Vis. Sat. range	PDOP range	Cost (Sats) range	Min. Vis. range
Improved GA (This study)	0.78 ± 0.02	0.12 ± 0.01	12.1–14.9	2.1–2.8	88–100	3.8–4.5
D-NSDE	0.72 ± 0.03*	0.18 ± 0.02*	11.5–14.0	2.3–3.2	85–100	3.5-4.0
PSO	0.65 ± 0.04*	0.25 ± 0.03*	10.8–13.2	2.7–3.6	80–100	3.0-3.8
SA	0.58 ± 0.05*	0.32 ± 0.04*	10.0-12.5	3.0–4.0	75–100	2.8–3.5

Table 11. Comparison of Multi-Objective optimization Performance. * indicates statistically significant differences compared to the improved GA ($p < 0.05$). Ranges represent the minimum–maximum values of solutions on the Pareto front for each objective.

Satellites	Algorithm	Vis. Sat.	PDOP	Time(s)	Mem(GB)
500	Improved GA (this study)	13.8	2.6	210	8.7
	D-NSDE	13.0	2.9	260	9.5
	PSO	11.8	3.3	190	6.5
1000	Improved GA (this study)	12.9	2.8	205	12.0
	D-NSDE	12.0	3.1	280	14.5
	PSO	10.5	3.6	220	8.2

Table 12. Scalability analysis (Different satellite Scales).

Algorithm variant	Vis. Sat.	PDOP	Time(s)	Conv. Gen.
Full GA (this study)	14.3	2.3	32	45
GA w/o Adapt.	13.1	2.7	38	55
GA w/o Parallel	13.5	2.5	185	45

Table 13. Ablation experiment Results.

3.0 after 10 years. The PSO-optimized constellation shows the greatest decline, with coverage falling to 87.1% and PDOP rising to 3.4. These results highlight the effectiveness of the improved GA's perturbation-resistant design, driven by features such as optimized relaxation factor γ. Regarding fault tolerance, the GA-optimized constellation loses only 3.5% points in coverage when a single satellite is randomly removed, dropping to 92.1%. Despite this loss, the average PDOP remains at 2.8, which is still suitable for high-precision positioning. In comparison, D-NSDE and PSO suffer more substantial coverage drops (− 4.8% and − 5.3%, respectively) and higher PDOP values (3.2 and 3.5). These findings suggest that the GA produces constellations with superior geometric redundancy and phase distribution, allowing adjacent satellites to more effectively compensate for isolated failures. Overall, these results confirm that the improved GA effectively integrates dynamic robustness into the constellation design. This enables systems to withstand long-term perturbations and unexpected failures. As a result, it ensures long-term stability and reliability for GNSS applications.

To further assess the GA's ability to manage complex multi-objective trade-offs—including visibility, PDOP, cost, and worst-case robustness—its Pareto front (derived using default weight settings) was compared against those from other algorithms. Two key performance metrics were used. The first is Hypervolume (HV), which measures the total space dominated by the Pareto front. Higher HV values indicate better solution quality and coverage. The second is Spacing (SP), which evaluates how uniformly the solutions are distributed. Lower SP values are better. A detailed comparison is presented in Table 11.

Table 11 clearly shows that the improved GA significantly outperforms all benchmark algorithms in MOO. It achieves the highest Hypervolume (0.78), indicating superior overall solution quality and broader coverage of high-performance solutions, compared to D-NSDE (0.72), PSO (0.65), and SA (0.58). Additionally, the improved GA yields the lowest Spacing value (0.12), reflecting a more uniform distribution of solutions along the Pareto front. This provides decision-makers with a richer and more balanced set of trade-offs. In terms of individual objective ranges, the improved GA consistently achieves broader or more favorable boundaries across all metrics. It delivers higher ranges for both the number of visible satellites (12.1–14.9) and minimum visible satellite count (3.8–4.5) than its counterparts. Its PDOP (2.1–2.8) and cost range (88–100 satellites) also outperform other methods. These results highlight the GA's ability to explore extreme edges of the objective space and identify solutions that either optimize single objectives or strike well-balanced trade-offs. For instance, the GA identifies configurations with only 88 satellites that still achieve a minimum visibility of 3.8 and a PDOP of 2.5. At the high-performance end, it finds solutions using 100 satellites that deliver 14.9 visible satellites and a PDOP of 2.1. Overall, the improved GA demonstrates excellent capability in balancing visibility, PDOP, cost, and worst-case robustness, offering system designers a Pareto front that is well-distributed, wide-ranging, and high in overall quality.

Scheme	Description	Avg. Vis.	Avg. PDOP	Min. Vis.	Cov.(%)	Sats
S2	High-inclination Walker + Polar + OCO	14.3	2.3	4.1	95.6	55
S1	Low-inclination Walker + Polar	13.6	2.5	3.8	93.5	60
S3	High-inclination Walker + OCO	13.9	2.4	3.9	94.2	51
S4	Single Polar Orbit	11.2	3.5	3.0	85.0	40
S5	Single OCO	10.5	3.8	2.8	80.5	30

Table 14. Performance comparison of constellation schemes optimized by improved GA.

Scheme	Equatorial coverage (%)	Equatorial Vis. Sat.	Std. Dev.
S2	**95.6**	**14.7**	**0.3**
S4	72.3	8.2	1.5

Table 15. Improvement of Equatorial coverage by OCO. Significant values are in bold.

To further assess the improved GA's scalability and the effectiveness of its core components, scalability tests and ablation studies were conducted (Tables 12 and 13). When the constellation size was increased to 500 satellites, the improved GA still achieved strong performance—averaging 13.8 visible satellites and a PDOP of 2.6—surpassing D-NSDE (13.0, 2.9) and PSO (11.8, 3.3). Although convergence time increased to 210 s, parallel processing kept it below D-NSDE's 260 s and only slightly above PSO's 190 s. Memory usage (8.7 GB) remained within acceptable limits. In the ultra-large-scale scenario with 1,000 satellites, the improved GA's advantages became even more pronounced. It achieved 12.9 visible satellites and a PDOP of 2.8—significantly better than D-NSDE (12.0, 3.1) and PSO (10.5, 3.6). Its convergence time (205 s), supported by 8-core parallelism, was well managed and considerably lower than the theoretical runtime for serial execution. Although memory usage increased to 12.0 GB, it can be alleviated through distributed computing. These findings confirm that the improved GA framework is well-suited for small- to medium-scale constellation design and, with parallelization, can effectively scale to ultra-large optimization problems.

Ablation experiments clearly validate the effectiveness of the two core enhancements: adaptive parameter tuning and parallel computing. Disabling the adaptive mechanism (GA w/o Adapt.) leads to a notable performance decline—visible satellites drop to 13.1, PDOP increases to 2.7, convergence time extends to 38 s, and the required generations rise to 55. These results emphasize the critical role of dynamic parameter adjustment in maintaining population diversity, preventing premature convergence, and accelerating convergence speed. In contrast, removing parallel computing (GA w/o Parallel) results in slightly better final performance than removing adaptation (13.5 visible satellites, PDOP of 2.5). However, convergence time drastically increases to 185 s—nearly six times longer—substantially undermining the algorithm's practicality. This highlights the importance of parallelization in efficiently managing the computationally intensive fitness evaluations. In summary, adaptive parameter tuning and parallel computing are both essential to the improved GA's strong performance and scalability. Together, they enable the framework to effectively tackle LEO constellation optimization problems across a wide range of problem sizes, from standard to ultra-large scale.

Finally, performance comparisons of five constellation schemes (S1–S5) optimized with the improved GA highlight the benefits of hybrid constellations, especially the role of OCO in enhancing equatorial coverage. Table 14 summarizes the performance of different schemes after optimization with the improved GA.

Analysis of Table 14 clearly demonstrates the superiority of hybrid constellations—particularly Scheme S2—and highlights the critical role of the Optimized Circular Orbit (OCO) component. Scheme S2, which integrates high-inclination Walker orbits (to optimize mid-latitude coverage), polar orbits (for high-latitude regions), and OCO (to enhance equatorial coverage), delivers the strongest overall performance. It achieves the highest or near-highest values across all key metrics: average visible satellites (14.3), coverage (95.6%), lowest average PDOP (2.3), and highest minimum visible satellites (4.1)—all with just 55 satellites. Schemes S1 (low-inclination Walker + polar) and S3 (high-inclination Walker + OCO) also perform well, outperforming the single-architecture schemes S4 (polar only) and S5 (OCO only). While S4 performs reasonably well at high latitudes, it fails to provide adequate coverage near the equator. In contrast, S5 is effective in equatorial regions but offers poor coverage at mid- and high-latitudes. These results strongly confirm the necessity of hybrid architectures to achieve globally balanced, high-performance coverage. Table 15 further illustrates the significant contribution of OCO to improving equatorial coverage.

Table 15 highlights OCO's exceptional effectiveness in addressing sparse equatorial coverage. In Scheme S2, the addition of OCO increases equatorial coverage from 72.3% (in the pure polar Scheme S4) to 95.6%—a substantial improvement of 23.3%. The average number of visible satellites at the equator rises from 8.2 to 14.7, while the standard deviation drops significantly from 1.5 to 0.3. This indicates not only denser but also more stable coverage. Based on these results, Scheme S2 is recommended as the optimal design in this study. Enabled by the improved GA, it delivers a high-performance, highly robust, and globally balanced hybrid LEO constellation. The inclusion of OCO plays a pivotal role in enhancing navigation capabilities, particularly in equatorial regions.

Discussion

The experimental results robustly demonstrate the overall superiority of the improved GA in solving complex LEO constellation optimization problems. The proposed method significantly outperforms PSO, SA, DP, and the latest D-NSDE algorithm across key navigation performance metrics, including visible satellite count, PDOP, and coverage. More critically, the optimized constellations exhibit strong dynamic robustness: they maintain high coverage and low PDOP under long-term orbital perturbations and show minimal performance degradation under single-satellite failures, highlighting excellent fault tolerance. These strengths arise from the algorithm's adaptive mechanisms, optimized relaxation factor (γ), and explicit integration of robustness metrics such as minimum satellite visibility. In MOO, the improved GA produces Pareto fronts that are uniformly distributed, broadly spanned, and superior in overall quality. This provides system designers with a diverse set of trade-off solutions balancing performance, cost, and robustness. Although its convergence time is slightly longer than that of PSO and SA, the significantly higher solution quality and the use of efficient parallel computation keep its computational overhead well within practical limits—even for large-scale scenarios. The experiments also underscore the benefits of hybrid constellation architectures—especially Scheme S2—over single-orbit configurations, and highlight the essential role of OCO in mitigating sparse coverage in equatorial regions. Ablation studies further confirm the critical importance of adaptive parameter tuning and parallel computing within the algorithmic framework. The proposed improved GA offers an effective and reliable approach for high-dimensional, dynamic, and multi-constrained LEO constellation design. The resulting solutions have strong potential to enhance the precision, reliability, and availability of future GNSS augmentation systems. Future research will explore the extension of this framework to mega-constellation optimization, autonomous online management, and integration with deep learning techniques.

Research in other fields also underscores the strong performance and versatility of GA. For instance, Squires et al. (2022) developed a novel GA-based approach for scheduling repetitive transcranial magnetic stimulation (RTMS) appointments. By implementing advanced survivor selection strategies and heuristic-based population initialization, their algorithm significantly improved the operational efficiency of medical centers. Experimental results showed that this GA-based method outperformed other scheduling techniques in reducing overall operational time[59]. In another study, Hamdia et al. (2021) applied GA-based integer-valued optimization to configure the architecture and features of machine learning models. Their method enhanced the performance of both deep neural networks and adaptive neuro-fuzzy inference systems. A case study in computational material design validated the effectiveness of their approach, highlighting GA's utility in improving model accuracy and configuration[60].

This study employs three optimization methods—GA, PSO, and SA, and SA—to enhance the visibility and positioning accuracy of LEO satellite constellations. Although GA shows strong performance, its application has certain limitations. First, the iterative nature of GA makes it challenging to meet the millisecond-level real-time decision-making requirements, such as emergency satellite attitude adjustments. In such cases, lightweight methods, such as rule-based approaches or RL, may be more appropriate. Second, when the number of optimization variables exceeds 20, the search space of GA expands exponentially, significantly increasing convergence time. In these situations, gradient-based algorithms, like Sequential Quadratic Programming, or surrogate-assisted evolutionary methods may be more efficient. Additionally, for convex optimization problems—such as fuel-efficient orbital transfers—traditional convex optimization techniques, such as interior-point methods, often outperform GA in both efficiency and accuracy. For example, Wang et al.[61] proposed that while GA excels in MOO problems, the interior-point method is 15 times faster and produces results closer to the optimal solution. Therefore, the applicability of GA should be assessed based on the problem structure, real-time requirements, and available computational resources. For complex, non-convex, multi-modal, and MOO tasks that allow for some computational delay—such as satellite constellation design—GA remains an ideal choice. In this study, an improved PSO method, as proposed by Kralfallah et al.[62], was applied to optimize the algorithm's parameters and accelerate convergence. Moreover, an adaptive adjustment mechanism was introduced into the SA algorithm, based on the hybrid strategy suggested by Long et al.[63], to further enhance its optimization performance. The integrated use of these methods not only increased optimization efficiency but also significantly reduced computational resource consumption. The experimental results show that GA outperformed the other methods in enhancing visibility and positioning accuracy, consistent with the findings of Wang et al. PSO demonstrated advantages in convergence speed, validating the effectiveness of Kralfallah et al.'s optimization algorithm. Furthermore, the SA algorithm, with the adaptive mechanism, showed significant improvements in resource utilization, aligning with the results reported by Long et al. These findings suggest that combining the latest optimization strategies can greatly enhance the performance of LEO satellite constellations.

This study presents a GA-based approach for optimizing the visibility of LEO satellite constellations, which could also be applied to MEO and GEO satellite constellations. However, due to the unique characteristics of MEO and GEO orbits, using GA for these constellations presents new challenges. First, MEO and GEO satellites differ from LEO satellites in terms of orbital altitude and relative motion speed. MEO and GEO satellites operate at higher altitudes, resulting in slower relative motion and less variation in coverage patterns. As a result, GA must carefully manage the relative positions of satellites and their coverage overlaps when optimizing constellations in these orbits. This challenge is particularly pronounced in GEO, where satellites remain fixed relative to the Earth's surface, unlike the rapid movement of LEO satellites. This fundamental difference requires GA to adapt its optimization strategy. Second, MEO and GEO constellations typically have fewer orbital planes and satellites. Therefore, the optimization process must place greater emphasis on how orbital plane configurations and phase factors affect global coverage. MEO satellites, with lower orbital inclinations, can provide broader coverage but often have fewer orbital planes, which may lead to visibility gaps in certain areas. In contrast, GEO satellites maintain nearly constant visibility from the ground, making their optimization more reliant on the strategic placement of satellites and phase factor adjustment. To address these unique challenges, GA parameters

may need to be adjusted based on the specific characteristics of MEO and GEO constellations, balancing computational cost with optimization effectiveness. Finally, optimizing MEO and GEO constellations involves high computational complexity, especially for large-scale satellite deployments and MOO scenarios. To mitigate this, parallel computing techniques can speed up fitness evaluation, and GA's crossover and mutation operations can be dynamically adjusted to fit the reduced number of orbital planes and satellites. By implementing these strategies, GA can be effectively applied to MEO and GEO satellite constellation optimization, improving the accuracy and reliability of satellite navigation systems.

Conclusion

Research contribution

This study proposes a novel adaptive parallel GA framework for the dynamic and robust design of hybrid LEO constellations. The key contributions are threefold. First, it introduces an optimization model that integrates polar, Walker, and OCO architectures. By explicitly encoding a dynamic relaxation factor (γ), the model incorporates a structural disturbance compensation mechanism, enabling resilient constellation designs against orbital perturbations. Second, an advanced GA framework is developed, featuring adaptive parameter tuning, parallel computing, and a structured fitness simplification strategy. This framework effectively solves multi-objective problems in high-dimensional mixed-parameter spaces, achieving a PDOP of 2.3 in the 100-satellite scenario and maintaining PDOP $\leq$ 2.8 in ultra-large-scale scenarios with up to 1,000 satellites. Third, a comprehensive robustness evaluation system is established. Experiments show that the optimized constellations experience only 66% of the coverage degradation seen in PSO-designed systems under single-satellite failure. Furthermore, the minimum number of visible satellites remains stable at 4.1 under perturbation, significantly improving system reliability. Overall, the proposed framework offers an engineering-ready tool for designing high-performance LEO-based GNSS augmentation constellations.

Future works and research limitations

Although this study focuses on satellite position calculation and constellation selection, it does not account for the impact of satellite dynamics and communication factors on system performance. The GA improvements, such as adaptive parameter adjustment and parallel computing optimization, have broad applicability to space missions. For instance, GA can dynamically adjust fuel consumption and orbital transfer time constraints in interplanetary trajectory optimization tasks, addressing multi-objective trade-offs. Furthermore, GA's global search capability can be applied to complex problems like deep-space network node deployment and asteroid resource exploration, showcasing its potential as a universal optimization tool for space missions. Future research could explore these factors to provide a more comprehensive evaluation of GNSS. Given the global nature of GNSS, future studies may also investigate strategies for achieving a global satellite constellation layout and integrating it with other navigation modes, such as inertial navigation and ground-based wireless positioning, to further enhance the performance and robustness of the navigation system.

Data availability

The datasets used and/or analyzed during the current study are available from the corresponding author Chao Qin on reasonable request via e-mail max_qinchao@hrbeu.edu.cn.

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Declarations

Competing interests

The authors declare no competing interests.

Ethics statement

This article does not contain any studies with human participants or animals performed by any of the authors. All methods were performed in accordance with relevant guidelines and regulations.

Additional information

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